Within a species, different races may show differences in susceptibility to infections. This is known as racial immunity. It has been reported that the African-descendants in the USA are more susceptible to tuberculosis than the Americans. The differences in innate immunity exhibited by different individuals in a race are known as individual immunity. It is well documented that homozygous twins exhibit similar degrees of resistance or susceptibility to lepromatous leprosy and tuberculosis.

10.3.1.2. Components involved in Innate Immunity

Components involved in innate immunity include skin, mucus, cells like neutrophils, macrophages, natural killer cells and soluble factors like complements, cytokines, and acute phase proteins.

Skin and mucus provide anatomical barrier, cells like macrophages and neutrophils provide phagocytic barrier and soluble factors like complement, acute phase proteins provides physiological and inflammatory barriers.

10.3.1.3 Mechanisms involved in Innate immunity

Innate immunity is provided by four types of defensive barriers namely, anatomical, physiological, phagocytic and inflammatory barriers.

Anatomical Barriers

Anatomic barriers that tend to prevent the entry of pathogens are an organism's first line of defense against infection. The skin and the surface of mucous membranes are included in this category because they are effective barriers to the entry of most microorganisms. The outer epidermis of skin contains several layers of tightly packed epithelial cells which prevents entry of pathogens. The outer epidermal layer consists of dead cells and is filled with a waterproofing protein called keratin. The inner dermis layer of skin, which is composed of connective tissue, contains blood vessels, hair follicles, sebaceous glands, and sweat glands. The sebaceous glands are associated with the hair follicles and produce an oily secretion called sebum. Sebum consists of lactic acid and fatty acids, which maintain the pH of the skin between 3 and 5; this pH inhibits the growth of most microorganisms.

The conjunctivae, the alimentary, respiratory, and urogenital tracts are lined by mucous membranes. These membranes consist of an outer epithelial layer and an underlying layer of connective tissue. The viscous fluid called mucus, which is secreted by epithelial cells of mucous membranes, entraps foreign microorganisms. In the lower respiratory tract, the mucous membrane is covered by cilia, the hair like protrusions of the epithelial cell membranes. The synchronous movement of cilia propels mucus-entrapped microorganisms from these tracts.

Physiological Barriers

The physiological barriers that contribute to innate immunity include temperature, pH, and various soluble and cell associated molecules. Many species are not susceptible

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